DAY 11

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DATE: 23-07-2025

Tasks:

1) Create an interface LibraryUser with the following methods declared,

Method Name

registerAccount

requestBook

2.Create 2 classes "KidUser" and "AdultUser" which implements the LibraryUser interface.

3.Both the classes should have two instance variables as specified below.

age	int
bookType	String

4.The methods in the KidUser class should perform the following logic.

1. registerAccount : if age < 12, a message displaying “You have successfully registered under a Kids Account” should be displayed in the console.

If(age>12), a message displaying, "Sorry, Age must be less than 12 to register as a kid" should be displayed in the console.

2. requestBook : if bookType is "Kids", a message displaying "Book Issued successfully, please return the book within 10 days" should be displayed in the console.

else, a message displaying, "You are allowed to take only kids books" should be displayed in the console.

5.The methods in the AdultUser class should perform the following logic.

1. registerAccount : if age > 12, a message displaying "You have successfully registered under an Adult Account" should be displayed in the console.

If age<12, a message displaying, "Sorry, Age must be greater than 12 to register as an adult" should be displayed in the console.

2. requestBook : if bookType is "Fiction", a message displaying "Book Issued successfully, please return the book within 7 days" should be displayed in the console.

else, a message displaying, "

You are allowed to take only adult Fiction books" should be displayed in the console.

6.Create a class LibraryInterfaceDemo with a main method which performs the below functions,

In the main method, test all the methods.

2) Write a program to read two integer array lists of size 5 each as input and to merge the two arrayLists, sort the merged arraylist in ascending order and fetch the elements at 2nd, 6th and 8th index into a new arrayList and return the final ArrayList.

3) Read student details as input.The details would include name, mark in the given order. The datatype for name is string, mark is float.Create a hashmap that contains name as key and mark as value.Get student name as input and display the student grade.

1. If Mark is less than 60, then grade is FAIL.

2. If Mark is greater than or equal to 60, then grade is PASS.

4) Write a program to get integers as input and store in the arraylist. Traverse the input list, if the number is even store in a arraylist called evenNumbersList and oddnumbers in oddNumberList. Print the input list and the lists containing even numbers and odd numbers.

Answers:

1. Code:

```
package org.learning1;
interface LibraryUser {
	void registerAccount();
	void requestBook();
}
class KidUser implements LibraryUser {
	int age;
```

```
	String bookType;
	KidUser(int age, String bookType) {
		this.age = age;
		this.bookType = bookType;
	}
	public void registerAccount() {
		if (age < 12) {
			System.out.println("You have successfully registered under a Kids
Account");
		} else {
			System.out.println("Sorry, Age must be less than 12 to register as a kid");
		}
	}
	public void requestBook() {
		if (bookType.equalsIgnoreCase("Kids")) {
			System.out.println("Book Issued successfully, please return the book
within 10 days");
		} else {
			System.out.println("You are allowed to take only kids books");
		}
	}
}
class AdultUser implements LibraryUser {
	int age;
	String bookType;
	AdultUser(int age, String bookType) {
		this.age = age;
		this.bookType = bookType;
	}
	public void registerAccount() {
		if (age > 12) {
			System.out.println("You have successfully registered under an Adult
Account");
		} else {
			System.out.println("Sorry, Age must be greater than 12 to register as an
adult");
		}
	}
	public void requestBook() {
		if (bookType.equalsIgnoreCase("Fiction")) {
			System.out.println("Book Issued successfully, please return the book
within 7 days");
		} else {
			System.out.println("You are allowed to take only adult Fiction books");
```

```java
			}
		}
}
public class LibraryInterfaceDemo {
	public static void main(String[] args) {
		System.out.println("*** Testing KidUser ***");
		KidUser kid1 = new KidUser(10, "Kids");
		kid1.registerAccount();
		kid1.requestBook();
		KidUser kid2 = new KidUser(14, "Fiction");
		kid2.registerAccount();
		kid2.requestBook();
		System.out.println("\n*** Testing AdultUser ***");
		AdultUser adult1 = new AdultUser(23, "Fiction");
		adult1.registerAccount();
		adult1.requestBook();
		AdultUser adult2 = new AdultUser(10, "Kids");
		adult2.registerAccount();
		adult2.requestBook();
	}
}
```

Output:

```
*** Testing KidUser ***
You have successfully registered under a Kids Account
Book Issued successfully, please return the book within 10 days
Sorry, Age must be less than 12 to register as a kid
You are allowed to take only kids books

*** Testing AdultUser ***
You have successfully registered under an Adult Account
Book Issued successfully, please return the book within 7 days
Sorry, Age must be greater than 12 to register as an adult
You are allowed to take only adult Fiction books
```

2. Code:

```java
package org.learning1;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Scanner;
public class MergeArray {
	public static void main(String[] args) {
		// TODO Auto-generated method stub
		Scanner sc = new Scanner(System.in);
		List<Integer> l1 = new ArrayList<>();
		List<Integer> l2 = new ArrayList<>();

		System.out.println("Enter 5 integers for List 1:");
		for(int i = 0; i < 5; i++) {
			l1.add(sc.nextInt());
		}

		System.out.println("Enter 5 integers for List 2:");
		for(int i = 0; i < 5; i++) {
			l2.add(sc.nextInt());
		}

		ArrayList<Integer> mergedList = new ArrayList<>(l1);
      mergedList.addAll(l2);
      Collections.sort(mergedList);
      System.out.println("Sorted Merged List: " + mergedList);
      ArrayList<Integer> finalList = new ArrayList<>();
      try {
         finalList.add(mergedList.get(2));
         finalList.add(mergedList.get(6));
         finalList.add(mergedList.get(8));
      } catch (IndexOutOfBoundsException e) {
         System.out.println("Index out of range. Make sure both input lists have 5 elements.");
      }
      System.out.println("Final List (elements at index 2, 6, 8): " + finalList);
      sc.close();
   }
}
```

Output:

Enter 5 integers for List 1:
6
4
8
2
3
Enter 5 integers for List 2:
1
5
7
4
9
Sorted Merged List: [1, 2, 3, 4, 4, 5, 6, 7, 8, 9]
Final List (elements at index 2, 6, 8): [3, 6, 8]

3. Code:

```
package org.learning1;
import java.util.*;
public class StudentGradeCheck {
	public static void main(String[] args) {
		Scanner sc = new Scanner(System.in);
		HashMap<String, Float> studentMap = new HashMap<>();
		System.out.print("Enter number of students: ");
		int n = sc.nextInt();
		sc.nextLine();
		for (int i = 0; i < n; i++) {
			System.out.print("Enter student name: ");
			String name = sc.nextLine();
			System.out.print("Enter mark for " + name + ": ");
			float mark = sc.nextFloat();
			sc.nextLine();
			studentMap.put(name, mark);
		}
		System.out.print("\nEnter the student name to check grade: ");
		String searchName = sc.nextLine();
		if (studentMap.containsKey(searchName)) {
			float mark = studentMap.get(searchName);
			if (mark >= 60) {
				System.out.println("Grade: PASS");
```

```
            } else {
                System.out.println("Grade: FAIL");
            }
        } else {
            System.out.println("Student not found.");
        }
        sc.close();
    }
}
```

Output:

Enter number of students: 3
Enter student name: sri
Enter mark for sri: 45
Enter student name: ram
Enter mark for ram: 68
Enter student name: kavi
Enter mark for kavi: 98
Enter the student name to check grade: kavi
Grade: PASS

4. Code:

```
package org.learning1;
import java.util.*;
public class EvenOddSeparator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Integer> inputList = new ArrayList<>();
        ArrayList<Integer> evenNumbersList = new ArrayList<>();
        ArrayList<Integer> oddNumbersList = new ArrayList<>();
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        System.out.println("Enter " + n + " integers:");
        for (int i = 0; i < n; i++) {
            int num = sc.nextInt();
            inputList.add(num);
            if (num % 2 == 0)
```

```
                    evenNumbersList.add(num);
                else
                    oddNumbersList.add(num);
            }
            System.out.println("\nInput List: " + inputList);
            System.out.println("Even Numbers List: " + evenNumbersList);
            System.out.println("Odd Numbers List: " + oddNumbersList);
            sc.close();
        }
}
```

Output:

```
Enter number of elements: 5
Enter 5 integers:
67
54
22
32
78
Input List: [67, 54, 22, 32, 78]
Even Numbers List: [54, 22, 32, 78]
Odd Numbers List: [67]
```